

Comparative Analysis of Airfoil Shapes at Transonic Speeds with Computational Fluid Dynamics

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Abstract— The shape of the airfoil is a critical factor in a plane wing's efficiency. However, each airfoil shape behaves differently depending on the flow regime. This study comparatively analyzes the effectiveness of different airfoil shapes based on three flow regimes: subsonic ($M = 0.5$), transonic ($M = 0.8, 1.2$), and supersonic ($M = 1.5$). Simulations were run using the Multiphase Reactive Flow Group's compressible flow code, HyBurn, written primarily in FORTRAN. Initial conditions were adjusted and the obstacle was embedded in the simulation as a '.STL' file, using the immersed boundary method. Simulation results revealed important information about the effectiveness of different aspects of airfoil shapes in the three flow regimes. It was found that the blunt nose of the airfoil performed well in subsonic cases, where it created significantly less drag than its counterpart, the sharp nose of the diamond airfoil. The flattened upper surface of the supercritical airfoil performed better in comparison to the conventional airfoil as it pushed the formation of the shockwave towards the aft part of the wing. The cambered rear section of the supercritical airfoil also performed better in terms of producing less drag than the other airfoils tested at transonic speeds. At supersonic speeds, the diamond airfoil performed better due to the sharpened nose, which causes a weaker oblique shock on the front surface of the wing. These results can be used in future studies in order to design more effective airfoils that could allow airplanes to travel closer to Mach 1.

Keywords—airfoil, transonic, subsonic, supersonic, camber, drag, lift, flow, Computational Fluid Dynamics

I. INTRODUCTION

As air travel continues to rise, the need for faster airplanes has led to increased research in the design of airfoils. As a result, the study of aerodynamics, specifically of the shape of the wing, or airfoil, on an airplane's performance has been the foundation of aerospace engineering for years. The airfoils developed by research in the past, such as the NACA 0012 airfoil, which features the conventional teardrop shape, have been sufficient for air travel at speeds below Mach 0.7, allowing for the planes to cruise at these speeds [3]. However, as humanity aspires to travel faster, new airfoil shapes are required in order for planes to travel safely.

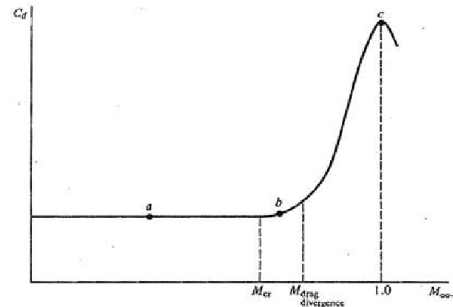


Figure I. Coefficient of Drag vs. Mach Number [7]

Transonic speeds, or speeds of Mach 0.8 - 1.2, present aircraft with a unique problem due to the formation of shock waves on the aircraft's wing. At these speeds, the airflow has no time to redirect its motion as it approaches the airplane's wing, causing it to get compressed by the leading edge of the wing and incoming air. This effect results in the formation of a shockwave. These shockwaves can be dangerous for aircraft because they split the airflow behind them, generating drag, as shown in Figure I, where the drag coefficient approaches a maximum as the Mach speed approaches 1. Given these shocks, the airfoil used on wings must be specifically designed in order to minimize the drag generated. At transonic speeds, increased drag may prove to be detrimental to the ability of the aircraft to fly [3]. Therefore, engineers must design airfoils that generate lift with the presence of shockwaves. Since shockwaves cause flow separation behind them, airfoils must be designed to push the shockwave to the back of the wing [2].

A key indicator of an airfoil's efficiency at high speeds is its critical speed, the Mach speed at which shocks first appear on the wing [3]. For the conventional airfoil, this speed is around Mach 0.74 [3]. Because most civilian aircraft travel at speeds of Mach 0.74 - 0.85, they are unable to use the conventional airfoil. As a result, engineers designed a new airfoil, called the supercritical airfoil. Unlike the conventional airfoil, supercritical airfoils feature a flattened upper surface and a curved aft section. Instead of trying to delay the shock, this design aims to push the shockwave towards the back of the wing and weaken the shock, reducing the loss of lift [3]. This design uses the flattened upper surface to slow the flow of air above the wing, giving the incoming airflow more time to move, thus delaying the onset of the shockwave. However, this design still has a critical Mach number well below the speed of sound, suggesting that there is still much room for improvement. While passenger jets such as the Concorde have traveled at speeds faster than Mach 1, they have also shown that traveling at supersonic speeds is inefficient and impractical. However, traveling at transonic speeds just under Mach 1 may prove to be a significant advancement in commercial aviation. Through the study of different airfoils, like the

modern supercritical airfoil, the goal is to find configurations that minimize drag and maximize the overall efficiency of the wing at transonic speeds, possibly allowing passenger planes to travel faster.

Understanding and optimizing airfoil shapes is critical to the development of faster, more efficient air travel for civilians. The current airfoil designs available allow civilian aircraft to travel at speeds up to Mach 0.85. However, new designs may allow aircraft to travel at speeds closer to the speed of sound, significantly reducing the time of travel.

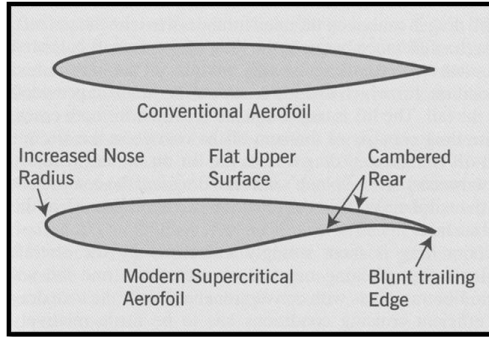


Figure II. Diagram of Conventional and Supercritical Airfoils [8]

The primary objective of this research is to evaluate the effectiveness of different airfoil shapes at transonic speeds, focusing on key metrics such as the lift and drag coefficients and pressure distribution as the configurations change. Through thorough examination of these parameters, the aim is to identify the most efficient airfoil configurations that can withstand the aerodynamic conditions at high-speed travel. Calculating values for the drag and lift coefficient as well as other values are incredibly costly through the use of physical experiments due to the heavy machinery, such as wind tunnels, required to model these flows. Instead, the use of computational methods makes running many simulations on aerodynamic processes significantly more cost and time effective. Computational Fluid Dynamics is the use of computers to predict liquid and gas flow. In order to do so, many calculations are computed based on the basic governing variables of compressible fluid flow. These simulations are then used to model airflow around various designs, allowing for precise measurements of multiple parameters.

Simulations make measuring numerous variables across multiple configurations much more efficient. As a result, we can design many configurations to test in order to properly establish correlations between different variables such as the speed and drag. With this data, the aim is to design a new configuration that under ideal conditions would allow planes to travel at speeds closer to Mach 1.

More efficient airfoils have implications in areas other than speed of travel. With the state of the climate at concern, ways to reduce emissions, especially in the aviation industry, are in need. More efficient airfoils could reduce the amount of fuel needed for travel and thus reduce the impact of the industry on the climate. Additionally, with more people using air travel, the safety of planes is an important consideration to make as planes travel faster. Simulations allow us to test configurations to ensure that they are safe to

apply in realistic situations. Finally, more efficient and faster airfoil designs may ultimately decrease the cost of air travel

II. METHODS

Given the variability of the location of the shockwave and the interaction of supersonic and subsonic flows in the given situation, using mathematical models to predict the flow can be time-consuming and error prone. Similarly, using physical apparatus in order to test aerodynamic bodies, such as wind tunnels, is incredibly costly. As a result, the primary method used to collect this data was Computational Fluid Dynamics (CFD). Through the use of CFD, we can measure the behavior of the airflow around the wing and simulate multiple airfoils and speeds without the use of costly and time-consuming experimental setups.

HyBurn is a program written primarily in FORTRAN developed by the Multiphase Reactive Flow Group at the University of Florida that uses CFD to model the flow of compressible fluids in different situations. Through the use of supercomputers, HyBurn completes numerous calculations on fluid dynamics, such as the Reimann problem, in order to generate simulations on the behavior of fluids in Multiphase Reactive Flow. Through the use of HyBurn, the lab's simulation code, we can input an airfoil shape as an STL file, run the simulation, and measure the values for the drag and lift as well as monitor the behavior of the shock on different airfoils and speeds. To do so, we started by feeding the simulation an airfoil shape and testing it at different speeds. The results were then compared in order to understand the efficiency of different configurations.

Computational Fluid Dynamics works by using the basic governing laws of fluid dynamics [1]. The foundational axioms of fluid dynamics dictate that mass, linear momentum, and energy are all conserved, yielding the equations [1]:

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u}{\partial x} = 0,$$

$$\frac{\partial \rho u}{\partial t} + \frac{\partial \rho u + p}{\partial x} = 0,$$

$$\frac{\partial \rho E}{\partial t} + \frac{\partial \rho E u + p u}{\partial x} = 0.$$

Given this, the values for three primitive variables, pressure, density, and velocity, can be calculated at each time step. The magnitude of a time step is determined by the formula:

$$\Delta t = \text{CFL} \frac{\Delta x}{\max_i (|u_i| + c_i)},$$

For these simulations, the Courant–Friedrichs–Lewy (CFL) stability criterion was 0.8. The primitive variables are then calculated recursively until the desired time is reached, 0.1 seconds for subsonic flow and 0.03 seconds for supersonic flow. This is done through the calculation of the flux of each conserved variable through a grid point at each time step. HyBurn uses the Godunov scheme in order to discretize the flux vectors:

$$\frac{\partial F}{\partial x} = \frac{F_{i+\frac{1}{2}}^n - F_{i-\frac{1}{2}}^n}{\Delta x}$$

Each conserved variable is then updated and used to calculate the primitive variable at each time step. From this, the primitive variables at each grid point are plotted and used to generate a simulation. Jobs are sent to the university's supercomputer, HiPerGator, in order to compute the numerous calculations in a reasonable time frame. Results are then stored and used to generate simulations such as the one shown below. The grid size was 9 x 6 meters for subsonic cases and 2 x 2 meters for supersonic cases.

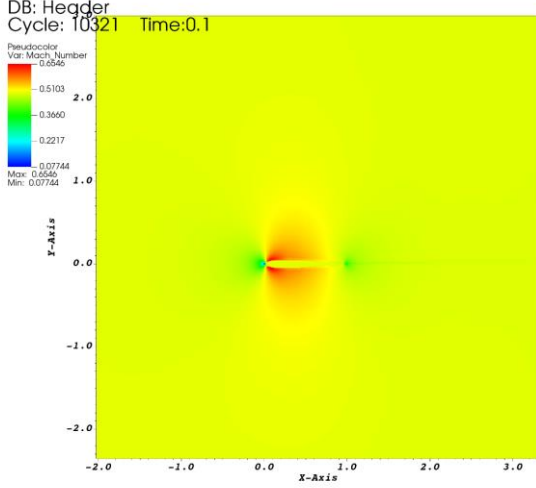


Figure III. HyBurn Simulation of Conventional Airfoil at Mach 0.5

HyBurn employs the immersed boundary method in order to allow the user to embed obstacles in the program. Given this, we can use airfoil shapes in the STL file format and place them in the simulation, as shown in the diagram above. After editing the initial conditions and boundary conditions, simulations for each airfoil shape are run: conventional (NACA 0012), supercritical (NASA SC(2)-0612), and diamond. For each of the shapes, four different initial conditions were run: Mach 0.5, 0.8, 1.2, 1.5. Data including the coefficients of drag and lift as well as other relevant information were monitored for each case. The Finite Volume Method was used to calculate the forces of drag and lift generated by each airfoil configuration. This employs Newton's second law in order to measure the sum of the forces at the surface of a finite volume through the use of integration. Finally, statistics and graphical methods were used to measure the correlation between configurations and efficiency. For each speed and airfoil, steady state flow will be confirmed through conservation of mass and momentum. By measuring the momentum flux through an interface, we can ensure that the simulation has reached a steady state, allowing us to obtain accurate data for analysis. To do so, a script written in Python measured the momentum and mass flux through a window. Results were compared to theoretical mathematical results to ensure the consistency of data compiled. For the conventional airfoil, the Thin Airfoil Theory was used to compare simulation results to experimental values. For the diamond airfoil, the Compressible Flow Theory was used.

III. RESULTS

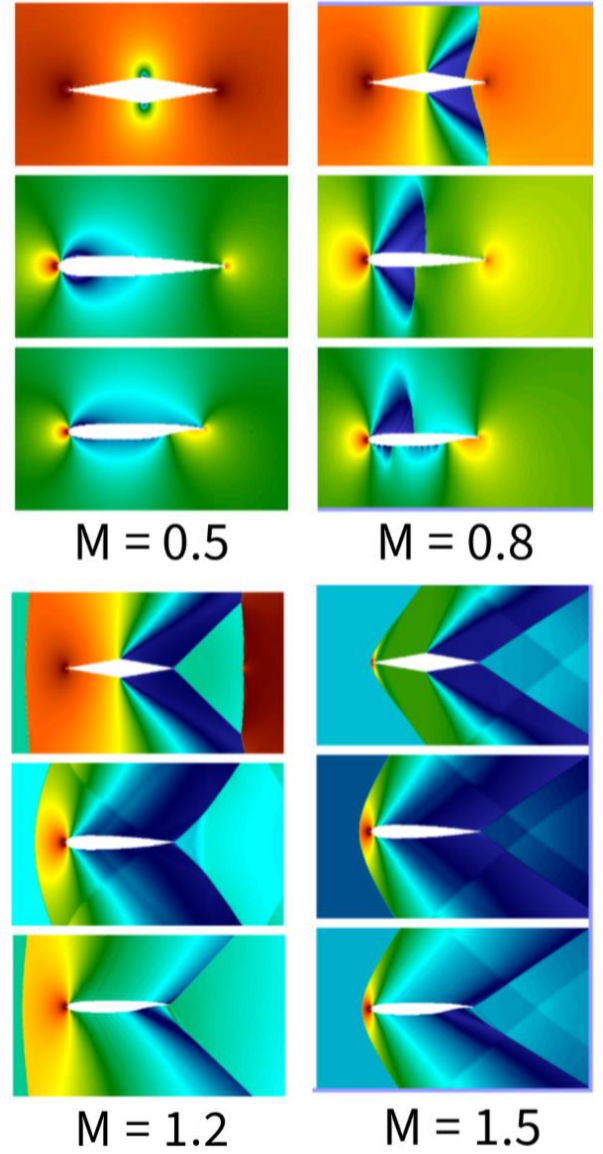


Figure IV. Simulation Results for Pressure Distributions of Airfoils at Flow Regimes at Mach Speed 'M'

Simulations were run for three different airfoil shapes in the order: diamond, conventional, and supercritical. They were each tested at four Mach speeds: 0.5, 0.8, 1.2, 1.5. For each simulation, warmer colors represent lower pressure, and the bulk air's pressure color is visible at the left side of the simulation. Upon confirming that the shockwaves closely follow expected results, a qualitative analysis was conducted on the efficiency of different aspects of the airfoils at different flow regimes.

IV. DISCUSSION

The drag generated is an important indicator of efficiency at different speeds. From a visual standpoint, it is clear that the conventional and supercritical airfoils perform better at subsonic speeds. This is because the sharp points on the surfaces of the diamond airfoil cause significant drag. Due to the sudden change in shape, the sharp point disrupts the airflow, causing drag, which can be seen with the green areas on the top and bottom surfaces of the diamond airfoil, which have accelerated airflow speeds of up to Mach 0.9. The curved surfaces of both the supercritical and conventional

airfoils allow the airflow to redirect in time, preventing the accumulation of air and drag. This is evident in the blue regions which closely follow the shape of the airfoils, indicating minimal flow separation.

At Mach 0.8 the diamond airfoil's sharp points cause significant flow separation, ultimately causing the formation of shockwaves that produce significant amounts of drag and minimize the amount of lift being produced. For this reason, the diamond airfoil provides little benefit at subsonic speeds. Instead, the supercritical airfoil has numerous benefits over the conventional airfoil at this speed due to its handling of the shockwaves. Due to its flatter top surface, the point of shockwave generation is pushed farther aft on the wing, making it more efficient from a drag point of view, as demonstrated by Figure IV, where the supercritical airfoil generates a weaker shockwave at Mach 0.8 compared to the conventional airfoil. The shockwave generated is weaker in comparison to the conventional airfoil, as shown by the steep drop off on the conventional airfoil compared to a rounded drop off on the supercritical airfoil. In addition, no shockwave is produced on the bottom surface of the supercritical airfoil, unlike the conventional one. This allows for lift to be produced in the presence of shockwaves, an important advantage to the supercritical airfoil. In general, the supercritical airfoil's ability to push the shockwave aft, as well as weaken it, makes it more efficient at transonic speeds.

At supersonic speeds, the involvement of shockwaves is inevitable. However, the diamond airfoil provides many benefits in comparison to the conventional and supercritical airfoils at these speeds. Due to its unique shape, the diamond airfoil is able to maintain high lift to drag ratios at more angles of attack. This is because unlike the subsonic airfoils such as the conventional and supercritical ones, the diamond airfoil features a sharp nose, creating weaker shockwaves. Despite the significant amount of drag produced at supersonic speeds, diamond airfoils excel at these speeds due to their ability to maintain high lift-to-drag ratios. At Mach 1.2, the conventional airfoil's blunt nose causes a larger area of higher-pressure flow at the leading edge of the wing, ultimately forming a strong bow wave, a type of shockwave formed at the front of the wing, as shown in Figure I. This bow wave disrupts the flow of air on both surfaces of the wing, causing significant drag and making the shape not suitable for supersonic flight. The supercritical airfoil also creates a bow wave at the front of the wing, however the flow separation is not as prominent as the conventional airfoil. This is because of the flattened shape of the foil, which allows the air to remain connected until the aft part of the wing, where the flow separation becomes apparent with the yellow and red regions, indicating faster airflow. For the conventional airflow, the separation of airflow is very prominent, resulting in the large area of faster airflow on the top and bottom surfaces of the wing.

At Mach 1.5, the airflow becomes so fast that any disruption in its flow results in a strong shockwave formed on the front surface of the wing. As shown in Figure IV, strong bow shocks are generated at the front of the conventional and supercritical airfoils. At supersonic speeds above $M = 1.2$, airfoils with blunt noses perform very poorly

due to the increased distance for the air to redirect to. As the speed increases, the diamond airfoil performs increasingly better due to many factors. The unique shape of the diamond causes an oblique shockwave to form on the front of the wing, however another expansion wave generates on the top surface of the wing as well. This shape allows for less drag due to the weaker bow shock formed, making it more viable than the subsonic airfoils. The diamond shape also provides structural integrity in comparison to the other airfoils tested.

V. CONCLUSION

The results reveal important information about the effectiveness of different aspect of airfoil shapes in their respective flow regimes. In subsonic flow, the blunt nose of the conventional and supercritical airfoils provides advantages in the minimization of drag and maximization of lift. However, they become less effective as the presence of shockwaves becomes more prominent. In addition, the flattened upper surface of the supercritical airfoil seems to be a significant factor in its effectiveness at transonic speeds due to its ability to push the shockwave towards the aft part of the wing. Compared to the conventional airfoil, the supercritical airfoil produces less drag at transonic speeds due to its cambered lower surface and flattened upper surface. At supersonic speeds, the sharp points of the diamond airfoil seem to provide benefits in terms of managing the presence of shockwaves. Since the desire is to travel at faster transonic speeds, future studies on the effectiveness of a hybrid between the supercritical and diamond airfoils could provide valuable insight into possible air travel at faster speeds than ever before.

VI. UNEXPECTED RESULTS AND LIMITATIONS

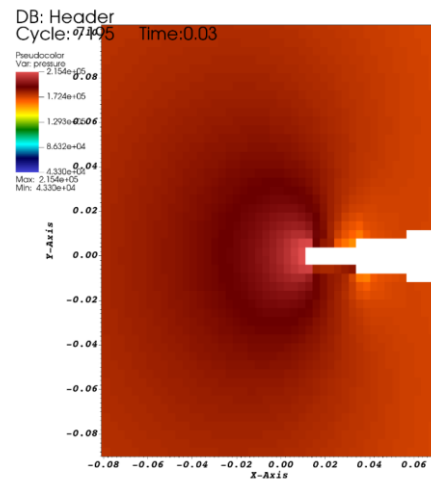


Figure V. Blunt Edge of Diamond Airfoil

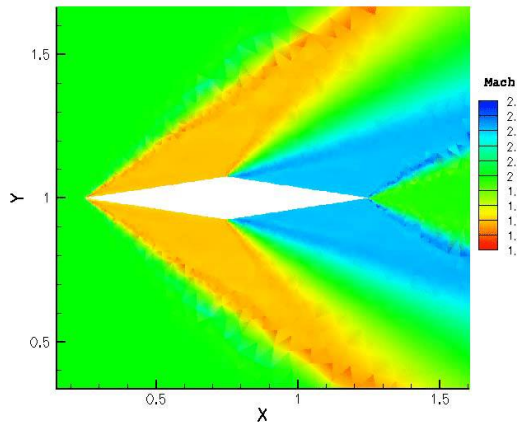


Figure VI. Desired Oblique Shockwave Shape [6]

It is important to note that for Mach 1.2, the expected result is not exactly obtained for the diamond airfoil due to a resolution error in the simulation. In the simulation, the front edge of the diamond airfoil is treated as a blunt shape, as shown in Figure V. While this error occurs in all flow regimes for the diamond airfoil, incorrect results are only visible for Mach 1.2 due to the nature of the shocks. The oblique shock formed on the diamond airfoil for the Mach 1.2 airflow behaves as a bow shockwave like the ones of the conventional and supercritical airfoils. Theoretically, the shockwave formed should take a diagonal shape, such as the one in Figure VI, rather than a rounded shape. This also explains why the bow shock in Figure IV for the diamond airfoil at Mach 1.5 takes on a rounded shape. As a result, it is important to resolve features of obstacles such as the leading edge of the diamond airfoil to ensure that the program does not assume a blunt point and thus result in a circular propagation of the shockwave. For this reason, any coefficients generated for the supersonic diamond airfoil cases are not comparable to analytic solutions. Due to the gridding technique employed by HyBurn, minor details such as the leading edge of the diamond airfoil are lost, causing errors like the one mentioned. In order to accurately measure the airfoils, a more resolved gridding was necessary.

There are also important limitations to this study in terms of implications and realistic applications. Some plane wings may have more than one airfoil being used on different parts of the wing. This can change the aerodynamic efficiency, making three-dimensional models necessary in order to accurately model the flow over the wing. In addition, most airplane wings have adjustable flaps in order to change the lift and drag being generated based on the desired outcome. This change could significantly alter the coefficients measured in this study, and must be considered before implementation. Finally, other factors such as the angle of attack of the wing can be important to the values measured in this study. In reality, traveling at an angle of attack of 0 degrees is unlikely, and different angles of attacks can change the formation of shockwaves and as a result the drag and lift being produced. Because this study was conducted over a 7-week period, the number of simulations that could be run were limited, and thus not all cases were considered in the study.

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